

Titanic - Machine Learning

(part 2)

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In [20]: # =====
# 1. DATA LOAD
# =====
import pandas as pd

url = "https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv"
df = pd.read_csv(url)

print(df)

df.columns = df.columns.str.strip()

df.columns = df.columns.str.lower()

print(df.columns)
# =====
# 2. CLEANING
# =====
df = df[['survived', 'pclass', 'age', 'sex', 'embarked']].head()

df['age'] = df['age'].fillna(df['age'].median())
df['sex'] = df['sex'].map({'male':0, 'female':1})
df['embarked'] = df['embarked'].fillna('S')
df['embarked'] = df['embarked'].map({'S':0, 'C':1, 'Q':2})

# =====
# 3. TRAIN TEST SPLIT
# =====
from sklearn.model_selection import train_test_split

trup=['pclass', 'age', 'sex', 'embarked']
X = df[trup]
y = df['survived']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

# =====
# 4. LOGISTIC REGRESSION
# =====
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

log_model = LogisticRegression()
log_model.fit(X_train, y_train)
log_pred = log_model.predict(X_test)

print("Logistic Accuracy:", accuracy_score(y_test, log_pred))

# =====
# 5. DECISION TREE
# =====
from sklearn.tree import DecisionTreeClassifier

tree_model = DecisionTreeClassifier()
tree_model.fit(X_train, y_train)
tree_pred = tree_model.predict(X_test)

print("Tree Accuracy:", accuracy_score(y_test, tree_pred))

# =====
# 6. SCALING
# =====
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2)

# =====
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# 7. LOGISTIC AFTER SCALING
# =====
log_model2 = LogisticRegression()
log_model2.fit(X_train, y_train)
log_pred2 = log_model2.predict(X_test)

print("Scaled Logistic Accuracy:", accuracy_score(y_test, log_pred2))

# =====
# 8. METRICS
# =====
from sklearn.metrics import precision_score, recall_score, f1_score

print("Precision:", precision_score(y_test, log_pred))
print("Recall:", recall_score(y_test, log_pred))
print("F1 Score:", f1_score(y_test, log_pred))
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	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	
..	...	...	...	
886	887	0	2	
887	888	1	1	
888	889	0	3	
889	890	1	1	
890	891	0	3	

	Name	Sex	Age	SibSp	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	
..	...	...	...	...	
886	Montvila, Rev. Juozas	male	27.0	0	
887	Graham, Miss. Margaret Edith	female	19.0	0	
888	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	
889	Behr, Mr. Karl Howell	male	26.0	0	
890	Dooley, Mr. Patrick	male	32.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S
..	...	...	...	...	...
886	0	211536	13.0000	NaN	S
887	0	112053	30.0000	B42	S
888	2	W./C. 6607	23.4500	NaN	S
889	0	111369	30.0000	C148	C
890	0	370376	7.7500	NaN	Q

```
[891 rows x 12 columns]
Index(['passengerid', 'survived', 'pclass', 'name', 'sex', 'age', 'sibsp',
      'parch', 'ticket', 'fare', 'cabin', 'embarked'],
      dtype='object')
Logistic Accuracy: 1.0
Tree Accuracy: 1.0
Scaled Logistic Accuracy: 1.0
Precision: 1.0
Recall: 1.0
F1 Score: 1.0
```

In []: